

INDICFDB: BENCHMARKING FULL-DUPLEX VOICE AGENTS ACROSS INDIAN LANGUAGES

Rajarshi Roy¹, Shobhit Banga², Jonathan Raiman¹, Supriya Paul², Bhaskar Singh²,
Manmeet Kaur², Sagar Jain², Hanuman Sidh², Pranav Sharma², Aditya Singh²,
Aaditya Pareek², Manas Dhir², Adi Margolin¹, Niket Agarwal¹, Bryan Catanzaro¹

¹NVIDIA

²VoiceArena

ABSTRACT

Full-duplex voice agents must handle pauses, take turns, backchannel, and respond to user interruptions in real time. Full-Duplex-Bench evaluates these behaviors, but its English-only corpus and reliance on word-timestamped ASR and an English-prompted LLM judge make it difficult to extend to Indian languages. We introduce IndicFDB, which extends it to ten languages spoken in India with 12,350 samples, nearly 17 times as many as the original. We address three challenges: finding conversational events in multilingual speech, evaluating their timing without reliable word-level alignment, and judging responses across languages. We mine pause handling, turn taking, and backchanneling samples from roughly 50,000 hours of channel-separated conversations using voice activity detection (VAD), and construct human-validated synthetic user interruption samples. Language-independent VAD heuristics evaluate timing, while an open-weight transcription and translation pipeline converts responses to English for LLM ratings of relevance and quality. Across seven voice agents, commercial APIs show unexpectedly consistent behavior across languages but are either fast or robust to pauses, never both, while monolingual open full-duplex models expose further tradeoffs among backchanneling, response quality, and latency.

Index Terms— full-duplex, voice agents, conversational dynamics, multilingual benchmark, Indic speech

1. INTRODUCTION

India, the world’s most populous country [1], is a major deployment setting for multilingual voice agents. Agents there must converse across languages with distinct scripts, pronunciations, and conversational conventions. Evaluating their real-time behavior is difficult because the automatic speech recognition (ASR), forced alignment, and semantic judging components that benchmarks rely on are unevenly supported across these languages. A benchmark built on English word timestamps or an English-only judge cannot simply be translated and scaled.

Full-Duplex-Bench (FDB) [2] evaluates four categories of conversational dynamics: pause handling, turn taking, backchanneling, and user interruption. FDB uses ASR with word-level timestamps to mine samples from human dialogue

and to evaluate agent timing, and prompts GPT-4o in English to judge the relevance and quality of agent responses to user interruptions. Ohashi et al. [3] instead use voice activity detection (VAD) to mine samples and define heuristic rewards that improve the same dynamics.

We introduce IndicFDB, a full-duplex benchmark covering ten languages spoken in India: Indian English, Hindi, Bengali, Punjabi, Gujarati, Marathi, Kannada, Telugu, Tamil, and Malayalam. We retain FDB’s four categories but build a far larger set of 12,350 samples (Table 1), using multilingual VAD [4] for both mining and evaluation. We also introduce a *transcribe-translate-judge* pipeline built entirely from open-weight ASR [5] and language [6] models, which rates user interruption responses in all ten languages. Evaluating four commercial APIs and three open models, we find that each commercial API behaves consistently across languages, yet no system is both fast and robust to pauses. The benchmark samples and associated mining and evaluation code are intended for release at <https://3bk47sbkf7b.github.io/indicfdb/>.

Table 1. Samples by conversational dynamics category.

Benchmark	Pause handling	Turn taking	Back channeling	User interruption	Total
Full-Duplex-Bench [2]	353	119	55	200	727
IndicFDB (per language)	315	235	385	300	1,235
IndicFDB (10 languages)	3,150	2,350	3,850	3,000	12,350

2. BENCHMARK CONSTRUCTION

2.1. Natural data curation

We automatically mine pause handling, backchanneling, and turn taking samples from VoiceArena’s corpus of channel-separated, two-speaker conversations, which contains roughly 5,000 hours of speech in each of the ten languages. Because

FDB’s word-level timestamping tools do not cover all ten languages [2], we rely on VAD-based heuristics as in [3]. Based on human validation feedback, we (i) refine the heuristics and (ii) add multi-threshold VAD filtering for high confidence.

Speech categorization: Silero VAD [4] segments each speaker’s channel into speech intervals called interpausal units (IPUs). We consider one speaker at a time, labeling that speaker X and the other speaker Y. Consecutive IPUs form utterances, split at silences longer than 0.5 s, and consecutive utterances form turns, split at silences longer than 1.5 s. Silences of 0.5–1.5 s are *short pauses*, and longer silences are *long pauses*. The mining heuristics for each category are as follows:

Pause handling: We select a single turn of speaker X, T_X , that satisfies three conditions: (i) $\text{duration}(T_X) \geq 8$ s; (ii) it contains at least one *short pause*; and (iii) speaker Y produces no speech.

Backchanneling: We select a single turn of speaker X, T_X , that satisfies two conditions: (i) $\text{duration}(T_X) \geq 15$ s; and (ii) it contains at least one short overlapping utterance (backchannel) U_{Y_k} from speaker Y with $\text{duration}(U_{Y_k}) \leq 1.2$ s.

Turn taking: We select turns T_X and T_Y from speakers X and Y that satisfy three conditions: (i) $\text{duration}(T_X) \geq 5$ s and $\text{duration}(T_Y) \geq 5$ s; (ii) $0 \leq \text{start}(T_Y) - \text{end}(T_X) \leq 0.5$ s; and (iii) no speech from Y during T_X or from X during T_Y .

Mined clips are filtered to avoid overlapping examples from the same source conversation and are then reviewed by language experts for validity.

Multi-threshold VAD filtering: At the default VAD threshold of 0.5, non-speech sounds could be detected as speech or real speech may go undetected; lowering the threshold worsens the former, and raising it worsens the latter. Running two VAD passes with low (0.1) and high (0.7) thresholds on a 1,000-hour multilingual corpus subset, we find 285,383 IPUs detected by both thresholds and an additional 25,971 IPUs (9.1%) detected only by the low threshold. The commonly detected IPUs are validated as speech, while the low-threshold-only IPUs include non-speech sounds and noisy, low-volume, or paralinguistic speech. We therefore mine only segments in which neither speaker has low-threshold-only IPUs, so that every detected IPU is speech and every silence is free of low-confidence speech.

2.2. Synthetic multilingual user interruption generation

We first expand FDB’s 200 English user interruption samples to 300. Each sample pairs a user *context* turn with an *interrupt* turn that cuts into the agent’s response to the context and should make the agent respond to the interruption instead. Language experts translate the *context* and *interrupt* turns into the ten languages with the aid of machine translation. Speech is synthesized with the Gemini 3.1 Flash TTS

Preview API [7]. Language experts validate every utterance, and the turns are stitched into a timed user interruption scenario following FDB.

3. MULTILINGUAL EVALUATION

As in FDB, we stream the user-side audio (speaker X) of each sample to each voice agent and record the agent’s output for the duration of the input. For mined samples, the other channel (speaker Y) serves as the ground-truth *Human* reference.

3.1. Conversational dynamics metrics

As in Section 2.1, we use VAD to segment agent speech into utterances and turns, labeling utterances shorter than 1.2 s as backchannels and utterances longer than 2 s as takeovers. FDB reports takeover rates, which are desirable in some categories and undesirable in others; we instead report a success rate for every category, so higher is always better. The metrics for each category are as follows:

Pause handling: A sample succeeds if the agent never takes over, i.e., it waits through the user’s short pauses.

Backchanneling: A sample succeeds if there are no takeovers. *Frequency* is the number of backchannels divided by the sample duration. *Jensen–Shannon distance (JSD)* compares agent backchannel timestamps with the ground-truth *Human* backchannel timestamps, using the same 0.2 s binning and smoothing as FDB. JSD is 1 when the agent produces no backchannels.

Turn taking: After the user’s turn T_U ends at $\text{end}(T_U)$, any agent turn T_A with $\text{duration}(T_A) > 1$ s and $\text{start}(T_A) > \text{end}(T_U)$ is labeled a response. A sample succeeds if a response exists and no takeover occurs before $\text{end}(T_U)$. Latency is $\text{start}(T_A) - \text{end}(T_U)$ for the earliest response T_A .

User interruption: Let T_U be the user’s interrupt turn. Agent IPUs overlapping $\text{start}(T_U)$ answer the *context* and are discarded; then any agent turn T_A with $\text{duration}(T_A) > 1$ s and $\text{start}(T_A) > \text{start}(T_U)$ is labeled a response. A sample succeeds if a response exists. Latency is $\max(0, \text{start}(T_A) - \text{end}(T_U))$.

Across samples in a category, success rate (S.R.) is the mean success value, and latency is the mean over successful samples; backchannel frequency and JSD are averaged over all samples, independent of success.

3.2. Response content evaluation

For successful user interruption samples, we transcribe the agent’s response audio with Bodhan Indic Transcribe Core [5], which supports all ten languages. Gemma 4 31B IT [6] translates non-English transcripts into English, while English transcripts pass through unchanged. Gemma 4 31B IT then rates each English response against English versions of the context and interrupt turns on FDB’s 0–5 relevance

scale. Judging in a single language keeps the judge prompt and rating scale identical across all ten languages. We chose these components after qualitatively inspecting errors from alternatives, including Gemma 4 E4B IT [8] for transcription and IndicTrans2 [9] and Bodhan Indic Translate [10] for translation.

4. RESULTS AND DISCUSSION

We evaluate four commercial voice APIs (GPT Live [11], Gemini 3.8 Live [12], Gemini 3.1 Flash Live Preview [13], and Grok Voice Think Fast 2.0 [14]) and three open full-duplex models (PersonaPlex [15], Moshi [16], and JoshTalks Human-1 [17]). Language coverage is unequal: GPT Live and Gemini 3.8 have outputs in ten languages, Gemini 3.1 in nine, Grok in three, PersonaPlex and Moshi in English, and Human-1 in Hindi. Table 2 therefore reports each available model-language pair rather than cross-language averages.

The commercial APIs occupy opposite ends of a latency–robustness tradeoff. GPT Live responds after 0.069–0.282 s on average, faster than the human reference in nine of ten languages, but often takes over when the user pauses: its success rate is 23.81–74.29% for pause handling and 15.32–43.90% for backchanneling. Gemini 3.8, Gemini 3.1, and Grok succeed on at least 99.05% of pause handling samples in every tested language, but respond after 1.125–1.400 s, 1.747–2.197 s, and 2.459–2.561 s, respectively. The same split holds for user interruptions, which GPT Live answers after 0.135–0.322 s and the other APIs after 1.098–1.946 s. No evaluated system escapes this tradeoff: across all 35 model-language pairs, every pair with sub-second turn taking latency has at most 90.16% pause handling success, and every pair above 99% responds after at least 1.125 s. TurnBench reports a similar speed–selectivity tradeoff for standalone turn-taking detectors [18].

Backchanneling success alone is insufficient, since silence never causes a takeover. Gemini 3.8 succeeds on 98.18–100% of backchanneling samples but produces only 0.009–0.027 backchannels/s against a human rate of 0.070–0.098/s, and Gemini 3.1 and Grok only 0.001–0.005/s; their JSD of 0.882–0.992 mostly reflects missing backchannels, which receive a JSD of 1. GPT Live matches or exceeds the human rate in every language while taking over in most of the same samples, and some of its sub-1.2 s utterances may be brief answers rather than backchannels. PersonaPlex comes closest to avoiding takeovers at a human-like rate, with 63.12% success at 0.091/s.

Although most of these languages have fewer public speech resources than English, each commercial API keeps the same behavioral profile across them. Gemini 3.8 succeeds on every pause handling sample in all ten languages with turn taking latency within 1.125–1.400 s, and the latency ordering GPT Live < Gemini 3.8 < Gemini 3.1 < Grok holds wherever these systems are compared. GPT Live’s success rates

vary more, e.g., from 23.81% (Bengali) to 74.29% (Tamil) for pause handling, but mined samples and human behavior also differ across languages (human backchannels: 0.070–0.098/s), so natural-sample metrics are best compared within a language. The parallel user interruption samples permit a more controlled comparison: Grok, Gemini 3.8, and Gemini 3.1 ratings stay within 4.787–4.997, whereas GPT Live falls from 4.980 in English to 4.550 in Punjabi and 4.003 in Gujarati. As Gemini 3.8 and 3.1 Gujarati responses pass through the same pipeline and receive 4.953 and 4.900, this gap more likely reflects GPT Live’s responses than the pipeline.

The open models show that timing and content metrics can rank systems differently. PersonaPlex is the only pair combining sub-second turn taking latency (0.360 s) with over 75% pause handling success (90.16%), and it has the lowest English JSD (0.799) with a 4.843 rating. Moshi takes over in 96.88% of backchanneling samples and has the lowest English rating (3.791). In Hindi, Human-1 exceeds GPT Live in pause handling (72.06 versus 61.59%) and backchanneling success (45.97 versus 15.32%) and has the lowest JSD (0.795), yet responds after 2.021 s with a 0.564 rating; since the same pipeline rates the other Hindi systems at 4.857–4.973, transcription and translation errors are unlikely to explain this gap fully.

5. CONCLUSION AND FUTURE WORK

IndicFDB makes full-duplex evaluation practical across ten languages spoken in India. Its 12,350 samples, nearly 17 times as many as FDB, are mined from roughly 50,000 hours of conversation with alignment-free, multi-threshold VAD heuristics and validated by language experts, and are scored by language-independent timing heuristics and a transcribe–translate–judge pipeline built entirely from open-weight models. Across seven voice agents, commercial APIs behave consistently across languages but favor either speed or pause robustness, and the best backchannel timing comes from open models that trail on robustness, latency, or response quality. No system yet combines human-like latency, pause robustness, natural backchanneling, and high-quality responses.

Two limitations remain: VAD heuristics may mislabel corner cases, such as brief answers as backchannels, and transcription and translation errors can accumulate before judging. Beyond evaluation, the corpus and mining heuristics could supply multilingual training data and rewards for interactivity alignment [3], and the mined samples could benchmark standalone turn-taking models in these languages, as TurnBench [18] does for end-of-turn and interruption detection. Releasing the samples and scripts will support these directions and extension to further languages.

Table 2. Per-language evaluation. S.R.: success rate (%); Lat.: latency (s); JSD: Jensen–Shannon distance; Freq.: backchannels/s; Rating: user interruption response rating (0–5). Human references have no user interruption output.

Language	Model	Pause handling	Turn taking		Backchanneling			User interruption		
		S.R.↑	S.R.↑	Lat.↓	S.R.↑	JSD↓	Freq.↑	S.R.↑	Lat.↓	Rating↑
English (Indian)	Human	100.00	100.00	0.273	100.00	0.000	0.070	–	–	–
	GPT Live	64.13	93.19	0.282	42.86	0.826	0.076	99.67	0.135	4.980
	Grok	99.05	92.34	2.459	98.70	0.989	0.002	100.00	1.739	4.923
	Gemini 3.8	100.00	100.00	1.338	99.48	0.953	0.009	100.00	1.157	4.973
	Gemini 3.1	100.00	99.15	1.747	98.96	0.981	0.003	100.00	1.485	4.997
	PersonaPlex	90.16	100.00	0.360	63.12	0.799	0.091	100.00	0.307	4.843
	Moshi	57.46	71.06	1.475	3.12	0.870	0.056	99.00	0.449	3.791
Hindi	Human	100.00	100.00	0.259	100.00	0.000	0.079	–	–	–
	GPT Live	61.59	91.06	0.181	15.32	0.822	0.109	100.00	0.322	4.857
	Grok	99.05	85.11	2.556	98.18	0.986	0.004	99.67	1.628	4.960
	Gemini 3.8	100.00	99.15	1.400	98.18	0.942	0.010	100.00	1.255	4.970
	Gemini 3.1	100.00	98.30	2.128	100.00	0.990	0.003	100.00	1.678	4.973
	Human-1	72.06	89.36	2.021	45.97	0.795	0.110	95.67	1.573	0.564
Bengali	Human	100.00	100.00	0.306	100.00	0.000	0.079	–	–	–
	GPT Live	23.81	100.00	0.189	37.92	0.801	0.115	100.00	0.216	4.853
	Grok	99.68	91.49	2.561	98.44	0.992	0.002	99.67	1.946	4.860
	Gemini 3.8	100.00	100.00	1.282	100.00	0.946	0.011	100.00	1.140	4.903
	Gemini 3.1	99.68	99.57	2.045	100.00	0.991	0.001	99.33	1.574	4.960
Punjabi	Human	100.00	100.00	0.245	100.00	0.000	0.095	–	–	–
	GPT Live	56.19	100.00	0.137	43.90	0.804	0.123	100.00	0.308	4.550
	Gemini 3.8	100.00	100.00	1.142	100.00	0.921	0.015	100.00	1.193	4.907
Gujarati	Human	100.00	100.00	0.289	100.00	0.000	0.089	–	–	–
	GPT Live	73.97	100.00	0.069	35.84	0.814	0.106	100.00	0.247	4.003
	Gemini 3.8	100.00	100.00	1.129	99.48	0.931	0.014	100.00	1.217	4.953
	Gemini 3.1	99.68	99.15	2.057	100.00	0.987	0.002	100.00	1.728	4.900
Marathi	Human	100.00	100.00	0.284	100.00	0.000	0.082	–	–	–
	GPT Live	54.29	100.00	0.107	37.14	0.808	0.128	100.00	0.260	4.850
	Gemini 3.8	100.00	100.00	1.161	100.00	0.947	0.012	100.00	1.158	4.973
	Gemini 3.1	100.00	100.00	1.883	99.22	0.981	0.003	100.00	1.563	4.973
Kannada	Human	100.00	100.00	0.321	100.00	0.000	0.098	–	–	–
	GPT Live	37.14	88.09	0.174	15.58	0.830	0.097	100.00	0.175	4.680
	Gemini 3.8	100.00	100.00	1.185	100.00	0.882	0.027	100.00	1.141	4.920
	Gemini 3.1	100.00	99.57	2.197	100.00	0.987	0.003	100.00	1.625	4.923
Telugu	Human	100.00	100.00	0.301	100.00	0.000	0.086	–	–	–
	GPT Live	61.90	99.57	0.090	41.56	0.797	0.116	100.00	0.168	4.837
	Gemini 3.8	100.00	99.15	1.190	99.74	0.898	0.019	100.00	1.098	4.877
	Gemini 3.1	100.00	98.72	1.884	99.74	0.976	0.005	100.00	1.578	4.980
Tamil	Human	100.00	100.00	0.311	100.00	0.000	0.081	–	–	–
	GPT Live	74.29	100.00	0.173	15.58	0.817	0.110	100.00	0.189	4.763
	Gemini 3.8	100.00	100.00	1.147	100.00	0.941	0.015	99.67	1.113	4.906
	Gemini 3.1	100.00	98.72	1.934	99.22	0.980	0.005	100.00	1.494	4.947
Malayalam	Human	100.00	100.00	0.289	100.00	0.000	0.087	–	–	–
	GPT Live	69.21	100.00	0.096	29.87	0.799	0.121	100.00	0.265	4.763
	Gemini 3.8	100.00	100.00	1.125	99.74	0.918	0.016	100.00	1.285	4.787
	Gemini 3.1	99.68	99.15	2.025	99.74	0.985	0.003	100.00	1.525	4.883

6. REFERENCES

- [1] United Nations Department of Economic and Social Affairs, “World population prospects 2024: Summary of results,” <https://desapublications.un.org/file/20847/download>, 2024.
- [2] Guan-Ting Lin, Jiachen Lian, Tingle Li, Qirui Wang, Gopala Anumanchipalli, Alexander H. Liu, and Hung-yi Lee, “Full-Duplex-Bench: A benchmark to evaluate full-duplex spoken dialogue models on turn-taking capabilities,” in *Proc. IEEE ASRU*, 2025, arXiv:2503.04721.
- [3] Atsumoto Ohashi, Neil Zeghidour, Alexandre Défossez, and Eugene Kharitonov, “Multi-faceted interactivity alignment in full-duplex speech models,” *arXiv preprint arXiv:2606.11167*, 2026.
- [4] Silero Team, “Silero VAD,” <https://github.com/snakers4/silero-vad>, 2024.
- [5] Bodhan AI, “Indic Transcribe Core,” <https://huggingface.co/bodhan-ai/indic-transcribe-core>, 2026.
- [6] Google DeepMind, “Gemma 4 31B IT,” <https://huggingface.co/google/gemma-4-31B-it>, 2026.
- [7] Google, “Gemini 3.1 Flash TTS (Text-to-Speech) Preview,” <https://ai.google.dev/gemini-api/docs/models/gemini-3.1-flash-tts-preview>, 2026, Model ID: gemini-3.1-flash-tts-preview. Accessed 2026-08-10.
- [8] Google DeepMind, “Gemma 4 E4B IT,” <https://huggingface.co/google/gemma-4-E4B-it>, 2026.
- [9] Jay Gala, Pranjal A Chitale, A K Raghavan, Varun Gumma, Sumanth Doddapaneni, Aswanth Kumar M, Janki Atul Nawale, Anupama Sujatha, Ratish Pudupully, Vivek Raghavan, Pratyush Kumar, Mitesh M Khapra, Raj Dabre, and Anoop Kunchukuttan, “IndicTrans2: Towards high-quality and accessible machine translation models for all 22 scheduled Indian languages,” *Transactions on Machine Learning Research*, 2023.
- [10] Bodhan AI and AI4Bharat, “Indic-Translate: Document-level machine translation for 22 Indian languages,” <https://bodhan.ai/research/blogs/indic-translate>, 2026.
- [11] OpenAI, “GPT-Live 1,” <https://developers.openai.com/api/docs/models/gpt-live-1>, 2026, Model ID: gpt-live-1. Accessed 2026-09-14.
- [12] Google, “Gemini 3.8 Live,” <https://ai.google.dev/gemini-api/docs/models/gemini-3.8-live>, 2026, Model ID: gemini-3.8-live. Accessed 2026-09-14.
- [13] Google, “Gemini 3.1 Flash Live Preview,” <https://ai.google.dev/gemini-api/docs/models/gemini-3.1-flash-live-preview>, 2026, Model ID: gemini-3.1-flash-live-preview. Accessed 2026-09-14.
- [14] xAI, “Grok Voice Think Fast 2.0,” <https://docs.x.ai/developers/model-capabilities/audio/speech-to-speech>, 2026, Model ID: grok-voice-think-fast-2.0. Accessed 2026-09-14.
- [15] Rajarshi Roy, Jonathan Raiman, Sang-gil Lee, Teodor-Dumitru Ene, Robert Kirby, Sungwon Kim, Jaehyeon Kim, and Bryan Catanzaro, “PersonaPlex: Voice and role control for full-duplex conversational speech models,” *arXiv preprint arXiv:2602.06053*, 2026.
- [16] Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour, “Moshi: A speech-text foundation model for real-time dialogue,” *arXiv preprint arXiv:2410.00037*, 2024.
- [17] Bhaskar Singh, Shobhit Banga, and Pranav Sharma, “Human-1: A full-duplex conversational speech model for Hindi,” Tech. Rep., JoshTalks, 2026.
- [18] Freeman Jiang et al., “TurnBench: A multi-domain benchmark for turn-taking dynamics in spoken dialogue,” *arXiv preprint arXiv:2608.25218*, 2026.